

PRISMA 2020 Checklist

Study: Compartment-Stratified Systematic Reanalysis of Complement, Coagulation, and Matrix Transcriptional Programs in Diabetic Kidney Disease

Item	PRISMA 2020 requirement	Location/reporting
1	Identify the report as a systematic review	Title identifies a systematic reanalysis
2	Report all PRISMA for Abstracts items	Abstract
3	Describe the rationale in context	Introduction, paragraphs 1–2
4	Provide an explicit objective/question	Introduction, final paragraph
5	Specify inclusion/exclusion criteria and synthesis grouping	Methods: Search, eligibility and selection
6	Specify all information sources and last search date	Methods: Search, eligibility and selection; Supplementary Tables S25–S27
7	Present complete search strategies	Source_Data_M20.zip/systematic_search
8	Describe selection methods, reviewers and automation	Methods: Search, eligibility and selection; Supplementary Table S22
9	Describe data collection and reviewer/automation process	Methods: Pathway family and expression processing; Reproducibility
10a	Define all outcomes sought	Introduction objectives; Methods: Gene synthesis and pathway inference
10b	Define other variables sought and assumptions for missing information	Table 1; Methods: Risk of bias; Supplementary Tables S2 and S21
11	Specify risk-of-bias assessment methods	Methods: Risk of bias, controls and ethics provenance
12	Specify effect measures	Methods: Pathway family and expression processing
13a	Describe processes for deciding study eligibility for each synthesis	Methods: Search/eligibility and Gene synthesis
13b	Describe data preparation methods	Methods: Pathway family and expression processing
13c	Describe tabulation and visual-display methods	Methods: Software; Figures 2–4; Supplementary tables
13d	Describe synthesis methods and rationale	Methods: Gene synthesis; Common-measurement pathway inference
13e	Describe methods to explore heterogeneity	Methods: Gene synthesis; Results: source sensitivity
13f	Describe sensitivity analyses	Results: Robustness; Supplementary Tables S6 and S10–S17
14	Describe assessment of reporting bias	Results: Robustness, bias and certainty boundaries
15	Describe certainty assessment	Results and Discussion; formal grading not performed and reason stated
16a	Describe results of the search and selection process	Results: Dataset identification; Figure 1
16b	Cite studies that appeared eligible but were excluded and explain why	Supplementary Table S1 lists every full-record exclusion and reason
17	Cite each included study and present its	Table 1; Supplementary Tables S2, S21 and S25;

Item	PRISMA 2020 requirement	Location/reporting
	characteristics	References 14–30
18	Present risk-of-bias assessments	Results: Robustness; Supplementary Table S21
19	Present individual-study summary statistics and effect estimates	Figure 4; Supplementary Tables S5–S9
20a	Summarize contributing studies' characteristics and risk of bias	Results; Table 1; Supplementary Table S21
20b	Present results of all statistical syntheses	Results; Figures 3–4; Supplementary Tables S4 and S6–S10
20c	Present investigations of heterogeneity	Gene meta-analysis heterogeneity fields; leave-one-source-out S10
20d	Present sensitivity-analysis results	Results: Robustness; Supplementary Tables S6 and S10–S17
21	Present assessment of reporting biases	Not estimable with three heterogeneous primary sources; stated in Results
22	Present certainty assessments	No formal grade; conditional language and limitations in Results/Discussion
23a	Provide a general interpretation in context	Discussion, paragraphs 1–4
23b	Discuss limitations of included evidence	Discussion, limitations paragraph
23c	Discuss limitations of review processes	Discussion, limitations paragraph
23d	Discuss implications for practice/policy/research	Discussion and Conclusion; no clinical recommendation made
24a	Provide registration information	Not registered; Introduction and Supplementary Table S24
24b	Indicate access to protocol	No prospective protocol; versioned specification history in Supplementary Table S24
24c	Describe and explain amendments	Supplementary Table S24; M20 explicitly labelled post-review
25	Describe support and funder role	Funding
26	Declare competing interests	Competing interests
27	Report availability of data, code and materials	Data availability; Code availability; Source_Data_M20.zip; Source_Code_M20.zip